

Raghad Sultan Alharbi

Data Engineering | Data Analytics | Data Governance

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ABOUT ME

Computer Science graduate working across the data lifecycle — from modelling and structuring data through to analysis and reporting. Enterprise experience building metadata infrastructure and data quality controls in Informatica CDGC, with project work in relational data modelling, ETL, machine learning pipelines and dashboard reporting using SQL, Python and Power BI.

EDUCATION

B.Sc. in Computer Science 
Umm Al Qura University

09/2022 – 06/2026 | Makkah, Saudi Arabia

PROFESSIONAL EXPERIENCE

Data Governance Intern (CO-OP), Master Works 

03/2025 – 07/2025 | Riyadh

- Built and maintained the centralised metadata layer for an enterprise data governance programme — a data catalog and business glossary in Informatica CDGC spanning multiple ERP and operational source systems.
- Ran automated metadata scans across source systems to profile and classify data assets, mapping sensitive and personally identifiable fields to support privacy controls and ownership assignment.
- Used SQL and Power BI to validate data, track classification and metadata coverage, and produce operational dashboards for programme reporting and progress tracking.

Tech Stack: Informatica CDGC, SQL, Power BI, Python, ETL

PROJECTS

Übär — Real-Time Multiplayer Web Application , *Graduation Project*

10/2025 – 06/2026

- Designed a real-time NoSQL architecture in Firebase for sessions, presence, and live updates.
- Engineered real-time data synchronization and concurrency-safe state management to maintain data consistency.
- Analysed usability data from 54+ participants using the System Usability Scale (SUS).

Tech Stack: JavaScript, Firebase Realtime Database, Node.js, NoSQL Data Modelling, Real-Time Data Processing

Video Game Sales Analysis & Prediction 

12/2024 – 01/2025

- Built an end-to-end data analysis pipeline covering data cleaning, EDA, and sales analysis across genre, platform, and publisher.
- Developed and evaluated a linear regression model using MSE and R^2 , with findings presented through an interactive dashboard.

Tech Stack: Python, Pandas, scikit-learn, Plotly, Streamlit

Hospital Database Management System 

09/2023 – 11/2023

- Designed and normalised a relational MySQL database with ERD, primary keys, foreign keys, and referential integrity.
- Developed analytical SQL queries using joins, aggregations, and grouping to analyse hospital operations.

Tech Stack: MySQL, SQL, Data Modelling, ERD

SMS Spam Detection — ML Pipeline 

10/2025 – 12/2025

- Built an end-to-end text classification pipeline with preprocessing, feature extraction, and Naive Bayes/SVM model comparison.
- Achieved 97% accuracy, 98% precision, and 92% F1, deploying predictions through Streamlit.

Tech Stack: Python, scikit-learn, NLTK, Pandas, Streamlit

Multi-Model Sentiment Analysis Framework — NLP 

04/2026 – 06/2026

- Built an NLP pipeline using BERT, RoBERTa, and DistilBERT for preprocessing, classification, and model evaluation.
- Compared models using Accuracy, Precision, Recall, and F1-score, with real-time predictions through Streamlit.

Tech Stack: Python, Hugging Face Transformers, scikit-learn, Pandas, Streamlit, NLP, Machine Learning

SmartSaver — Personal Finance Management System 

05/2025 – 03/2025

- Designed a relational MySQL database for users, salaries, expenses, and financial records.
- Developed logic to calculate savings rates, spending categories, and real-time financial metrics for dashboard reporting.

Tech Stack: PHP, MySQL, SQL, JavaScript, HTML, CSS, Data Modelling, Agile/Scrum

Aljamoum University Campus Network Design 

12/2024 – 01/2025

- Designed a scalable campus network with VLAN segmentation for wired and wireless users.
- Configured routing and DHCP and performed connectivity testing to validate network performance.

Tech Stack: Cisco Packet Tracer, Routing & Switching, VLANs, DHCP, Static Routing

SKILLS

Data Engineering & Databases

SQL, MySQL, Data Modelling, ETL, Data Warehousing, Firebase Realtime Database

Data Governance

Informatica CDGC, Data Catalog, Business Glossary, Metadata Management, Data Classification, Data Quality

Coursework Exposure

Apache Spark, PySpark, Big Data Concepts, Cloud Computing (Azure IaaS)

Languages

Arabic, English

Data Analysis & BI

Pandas, NumPy, scikit-learn, Power BI, Plotly, Streamlit, Data Profiling

Programming & Tools

Python, SQL, JavaScript, Git, GitHub, Agile/Scrum

Soft Skills

Problem Solving, Communication, Teamwork, Time Management